

Week 3

Commercial Banks- Part I

Banks' Role in Money Creation

Readings for the week:

- Pages 49-52 & 63-75 in chapter 3 of "Financial Institutions and Markets" by Howells & Bain, 2007 (on the reading list)
- McLeay, M., Radia, A. and Thomas, R., 2014. Money creation in the modern economy. Bank of England Quarterly Bulletin, 2014 Q1, pp.14-27 (on BB older for Week 3)
- "How Banks Create Money" at Positivemoney webpage (on reading list)

A Word of Warning About Bank Classifications

Since 1990s, financial institutions diversified their services.

- The divisions among commercial, investment, retail, wholesale banks are not quite clear.

Define these as activities instead of types of institutions

Royal Bank of Scotland Group in 2006

Retail banking	Citizens	RBS Insurance	Direct Retail	Manufacturing	Wealth management	Ulster Bank	Corporate banking and financial markets
Banking, insurance and financial services for individuals and small businesses	Retail and corporate banking activities through its branch network in 13 states in the northeastern US and through non-branch offices in other states.	Sells and underwrites retail, commercial and wholesale insurance on the telephone, the internet, and through brokers and intermediaries.	Retail and wholesale insurance via phone and the internet	Technology support and payment mechanisms	Private banking and offshore banking services	Retail and wholesale banking services to clients in Northern Ireland	Banking services to firms, derivatives, forex money market, bond and gilt market making
UK financial services	US banking	Insurance	Internet banking	Support services	Banking services for overseas	NI financial services	Corporate and investment banking

Source: Howells & Bain "Financial Institutions and Markets"

A Taxonomy of Banking in the UK

	Retail banking		Wholesale banking	
	Personal	Small business	Corporate	Investment
Retail banks				
– the 'Big Five'	●	●	●	●
– others	●	●	●	●
Wholesale banks				
– UK 'merchant banks'			●	●
– Foreign banks			●	●

Key: ● = heavily involved; ● = limited involvement

Source: Howells & Bain "Financial Institutions and Markets"

Retail banking is dominated by the so-called 'Big Five' in the UK:

HSBC, Barclays, Lloyds, NatWest, and Santander

Topics

- Bank of England
- Different Forms of Money
- Liquidity and Various Definitions of Money Supply
- Typical Bank Balance Sheet
- Money creation – How is money created.

Bank of England

The UK's central bank, (the equivalent is the Fed in the USA)

Typical responsibilities

- the conduct of monetary policy (<https://www.bankofengland.co.uk/monetary-policy>)
- banker to the commercial banking system
- banker to the government - responsible for managing all the account transactions of the Government
- supervisor of the banking system
- manager of the foreign exchange reserves
- issuer of the national currency.

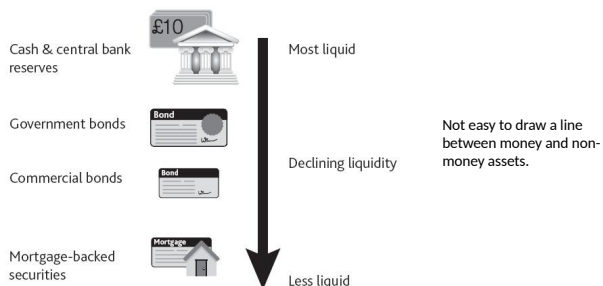
Different Forms on Money

1. Cash (banknotes and coins)
 - Created under the authority of Bank of England
 - Around 3% of the circulating money supply
2. Electronic money
 - Bank of England creates electronic money for the banking system
 - Banks have their own accounts with Bank of England
 - Money in these accounts are called central bank reserves
 - Banks create electronic money for their customers
 - By issuing loans to customers.

Liquidity and Near-Money

Liquidity: The speed and convenience with which an asset can be converted into money

- Some assets are more liquid than the others



Measuring Money Supply

- How do we measure money? Which particular assets can be called money ?
- Monetary aggregates is constructed using the concept of liquidity.

Bank of England's Definitions of Money Supply

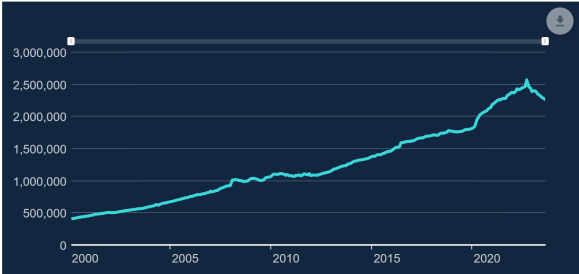
Narrow money (also known as M0, high-powered money, monetary base, or central bank money)

1. Notes and coins in circulation (i.e. **Cash**)
2. Reserve balances at the Bank of England

Broad Money

- M1: **Cash** plus current accounts
- M2: M1 plus savings accounts
- M3: M2 plus repurchase agreements, money market funds, and debt securities up to two years (not commonly used anymore)
- M4: M3 plus other deposits at UK banks and building societies.

Monthly Amounts Outstanding of M1 (in £ millions) Seasonally Adjusted



Source: Bank of England

Typical bank balance sheet

The Balance Sheet is a list of an entity's assets and liabilities.

ASSETS	LIABILITIES
Loans	Deposits
Mortgages	
Consumer Loans	
Lending to companies	
Other Assets	Shareholders' equity
Real estate	
Bank accounts	

A bank's balance sheet lists sources of bank funds (liabilities) and uses to which they are put (assets)

Total assets = Total liabilities + Capital (equity)

Balance Sheet of All Commercial Banks in the US in 2016

ASSETS		LIABILITIES	
Reserves and cash items	16%	Current accounts	10%
Securities	20%	Savings accounts (i.e. time-deposits)	60%
Loans	55%	Borrowings	19%
Other Assets (e.g., buildings)	9%	Bank Capital (equity)	11%
TOTAL	100%	TOTAL	100%

Liabilities

ASSETS		LIABILITIES	
Reserves and cash items	16%	Current accounts	10%
Securities	20%	Savings accounts (i.e. time-deposits)	60%
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Other Assets (e.g., buildings)	9%	Bank Capital (equity)	11%
TOTAL	100%	TOTAL	100%

Current accounts are the lowest cost funds for the bank –

- safe and liquid, but offer low interest to customers.

Savings accounts are the primary source of funds

- Highest cost of funding, but most stable for bank

Borrowings: funds from the central bank, other banks, and corporations

- More volatile than other liabilities

Bank Capital: is the source of funds supplied by the bank owners.

Assets

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Other Assets (e.g., buildings)	9%	Bank Capital (equity)	11%
TOTAL	100%	TOTAL	100%

Reserves are funds held in account with the central banks

Cash items are

1. funds from other banks that have not yet been transferred
2. Deposits at Other Banks

Securities are typically government securities

Loans: a bank's income-earning assets, not very liquid

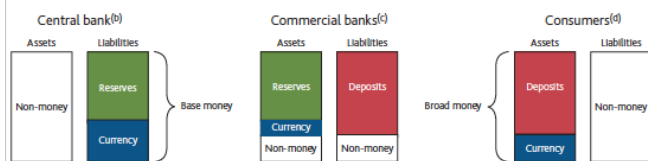
- Typically include business loans, auto loans, and mortgages

Other Assets: bank buildings, computer systems, and other equipment.

Lloyds Balance Sheet

Summary balance sheet	At 31 Dec 2022 £m	At 31 Dec 2021 £m	Change %
Assets			
Cash and balances at central banks	91,388	76,420	20
Financial assets at fair value through profit or loss	180,609	206,771	(13)
Derivative financial instruments	24,753	22,051	12
Financial assets at amortised cost	520,322	517,156	1
Financial assets at fair value through other comprehensive income	23,154	28,137	(18)
Other assets	37,603	35,990	4
Total assets	877,829	886,525	(1)
Liabilities			
Deposits from banks	7,266	7,647	(5)
Customer deposits	475,331	476,344	
Repurchase agreements at amortised cost ¹	48,596	31,125	56
Financial liabilities at fair value through profit or loss	17,755	23,123	(23)
Derivative financial instruments	24,042	18,060	33
Debt securities in issue	73,819	71,552	3
Liabilities arising from insurance and investment contracts	149,868	168,463	(11)
Other liabilities	22,901	23,951	(4)
Subordinated liabilities	10,730	13,108	(18)
Total liabilities	830,308	833,373	
Total equity	47,521	53,152	(11)
Total equity and liabilities	877,829	886,525	(1)

Figure 2 Stylised balance sheets of different types of money holders and issuers in the economy^(a)



Two misconceptions about money creation

#1: “Banks act simply as intermediaries, lending out the deposits that savers place with them”

In reality, commercial banks are the **creators** of deposit money

#2: Only central bank determines the quantity of money in the economy

Partially true. Rather than controlling the quantity of money, central banks today typically try to set the price of money — that is, interest rates

[THIS VIDEO EXPLAINS IMPORTANT MISCONCEPTIONS ABOUT BANKING AND MONEY](https://www.youtube.com/watch?feature=shared&v=bE8i-4HpKIM)

(<https://www.youtube.com/watch?feature=shared&v=bE8i-4HpKIM>).

Money Multiplier Model

Bank A Balance Sheet			
Assets		Liabilities + Net Worth	
Reserves	\$10 million	Deposits	\$10 million

Assets		Liabilities + Net Worth	
Reserves	\$1 million	Deposits	\$10 million
Loan to Hank's Auto Supply	\$9 million		

Money Multiplier Model

- Suppose Hank deposits the loan in his regular account with Bank B
- The deposits at Bank B rise by \$9 million and its reserves also rise by \$9 million

Assets		Liabilities + Net Worth	
Reserves	+ \$9 million	Deposits	+ \$9 million

Now, Bank B must hold some required reserves (\$900,000) but can lend out the other amount (\$8.1 million) in a loan to Jack's Chevy Dealership.

Assets		Liabilities + Net Worth	
Reserves	\$90,000	Deposits	+ \$9 million
Loans	\$8.1 million		

Money Multiplier Model

Assets		Liabilities + Net Worth	
Reserves	+ \$8.1 million	Deposits	+ \$8.1 million

- If Jack's Chevy Dealership deposits the loan in its account at Bank C, the money supply just increased by an *additional \$8.1 million*
- Making loans that are then deposited into a deposit account increases the money in circulation
 - Banks are required to hold only a fraction of their deposits
 - loans end up deposited back into banking system
 - which increases deposits and the money supply.

Money Multiplier Model

- Money multiplier model is an incomplete depiction of how money is created in a modern economy
- There is no reserve requirement ratio for banks in the UK

[THIS VIDEO EXPLAINS HOW MONEY MULTIPLIER MODEL WORKS \(OR DOES NOT WORK\)](https://www.youtube.com/watch?feature=shared&v=SkAzDrrKkME)

<https://www.youtube.com/watch?feature=shared&v=SkAzDrrKkME>
(the first 3 minutes is the key).

Money Creation

[THIS VIDEO EXPLAINS HOW BANKS CREATE MONEY](https://www.youtube.com/watch?v=KvpbQlQwl0A&t=395s)

[https://www.youtube.com/watch?
v=KvpbQlQwl0A&t=395s](https://www.youtube.com/watch?v=KvpbQlQwl0A&t=395s)

(from 6 min 35 sec – 9 min 35 sec)

- Banks create money every time they make a loan
- Loans create deposits.

Money Creation

[THIS VIDEO EXPLAINS HOW SIMPLE INTERBANK PAYMENTS HAPPEN](https://www.youtube.com/watch?v=KvpbQlQwl0A&t=577s)

<https://www.youtube.com/watch?v=KvpbQlQwl0A&t=577s>
(from 9 min 37 sec – 12 min 45 sec)

[THIS VIDEO EXPLAINS HOW INTERBANK PAYMENTS HAPPEN WITH MULTIPLE BANKS](https://www.youtube.com/watch?v=KvpbQlQwl0A&t=767s)

<https://www.youtube.com/watch?v=KvpbQlQwl0A&t=767s>
(from 12 min 47 sec – 15 min).

Thank you
